

Sales Dashboard Design

Veda Technologies – Data Analytics Internship
Task: Interactive Sales Performance Dashboard

This report documents the development of an interactive sales dashboard created to transform cleaned sales data into a stakeholder-friendly analytical view. The dashboard was developed using Microsoft Power BI and focuses on sales performance across products, regions, and time.

1. Executive Summary

The objective of this task was to design a basic interactive dashboard that allows a stakeholder to understand sales performance without manually examining raw data. The dashboard brings together key business measures and visual analysis into a single interactive reporting experience. The design follows the task requirements of using a limited number of meaningful KPIs, multiple visuals, and slicers/filters that allow users to explore the information according to their needs.

The completed Power BI work focuses on three major analytical dimensions: product performance, regional performance, and sales performance over time. Interactive filtering is incorporated so that the user can change the analytical context and observe the dashboard respond to the selected filters.

2. Task Objective

The primary objective is to convert cleaned sales data and business KPIs into an interactive dashboard that a stakeholder could realistically use for monitoring and understanding sales performance.

- Present important sales KPIs in a concise format.
- Compare sales performance across products.
- Compare sales performance across regions.
- Analyze sales performance across different periods.
- Provide interactive filtering through slicers/filters.
- Create a dashboard that is simple enough for quick decision-making.

3. Tools and Technologies

Tool	Purpose
Microsoft Power BI	Dashboard development, data modeling, visuals, filters and interactive reporting
Excel / Source Data	Source and/or prepared sales data used for analysis
Power BI Visuals	KPI cards, charts, tables and interactive visual elements

4. Data Preparation and Modeling

Before creating the dashboard, the sales data needs to be structured so that Power BI can correctly aggregate and visualize the measures. Important fields such as product, region, date/month, sales and other relevant business attributes are used as dimensions or measures. Date-related fields support the time

analysis, while product and region fields support categorical comparisons.

The dashboard workflow follows the general analytics process: prepare the data → load it into Power BI → verify fields and data types → create measures/aggregations → build visuals → add slicers → format the dashboard → validate interactions.

5. KPI Design

The task recommends limiting a first dashboard to approximately three or four KPIs. The dashboard design therefore prioritizes high-level measures that can be understood quickly before the user examines detailed charts.

- Total Sales – provides an overall view of revenue generated in the selected context.
- Profit / Profit-related measure – helps evaluate financial performance alongside Sales.
- Units/Quantity Sold – provides a volume perspective where the dataset supports it.
- Regional or other high-level performance indicator – provides quick comparison between business areas.

Note: The exact displayed KPI values depend on the data loaded into the completed Power BI file and the filters selected by the viewer.

6. Dashboard Visualizations

6.1 Sales by Product

The product-level visual compares sales across individual products or product groups. Its purpose is to help a stakeholder identify differences in sales contribution and understand which products require attention. A product comparison is more useful than displaying a raw product column because it directly supports a business question: how is sales performance distributed across products?

6.2 Sales by Region

The regional visual compares sales performance across geographic regions. This allows stakeholders to identify differences between markets and investigate regional performance using the available filters.

6.3 Sales by Time / Year

The time-based visual presents sales across the available time period. It supports trend analysis by showing how sales change over time and makes periods of higher or lower activity easier to identify.

6.4 Supporting KPI / Summary Visuals

KPI cards and supporting summary elements provide a high-level snapshot before the viewer moves into the detailed product, regional, and time-based visuals. This creates a top-down reading flow: overall performance first, followed by detailed comparisons.

7. Interactivity and Filters

Interactivity is a core requirement of the dashboard. Slicers/filters allow users to change the analytical context without rebuilding the report. The dashboard uses filtering concepts such as month/time and region so that a stakeholder can focus on a particular period or geographic area.

- Select a region to view performance for that region.
- Select a time period/month to examine sales during a specific period.
- Combine filters to narrow the analysis to a particular region and period.

- Observe the connected visuals update according to the active selection.

8. Dashboard Design Principles

- Keep the number of primary KPIs limited so the dashboard remains easy to scan.
- Use consistent formatting and terminology across all visuals.
- Use clear chart titles that communicate what is being measured.
- Maintain a logical visual hierarchy from KPIs to detailed charts.
- Avoid unnecessary charts that do not answer a specific business question.
- Use consistent color coding where color communicates a meaningful comparison.
- Keep filters visible and easy to understand.

9. One-Page Dashboard Reading Guide

Step 1 – Start with the KPI section: Review the headline sales/performance measures to understand the overall state of the selected data.

Step 2 – Check the time trend: Look for increases, decreases, peaks, or periods requiring further investigation.

Step 3 – Compare regions: Use the regional visual to understand geographic differences.

Step 4 – Compare products: Examine product-level performance to understand where sales are concentrated.

Step 5 – Apply filters: Select a month/time period or region and observe how the dashboard changes. This turns the dashboard from a static report into an interactive analytical tool.

10. Business Value

The dashboard converts transactional sales information into a visual reporting layer that can support routine performance monitoring. A stakeholder can quickly compare products, regions, and time periods and then use filters to investigate a specific area. This reduces the need to manually inspect raw data and supports faster exploratory analysis.

11. Interview Questions and Answers

How do you decide which KPIs belong on a dashboard versus a detailed report?

KPIs should represent the most important business questions and provide a quick overview of performance. Detailed reports can contain supporting metrics and granular records that are not necessary for an executive snapshot.

What makes a dashboard interactive rather than just a static chart?

Interactivity allows the user to change the context of the analysis through slicers, filters, drill-downs, cross-filtering, or similar controls, with connected visuals responding to those actions.

How would you design a dashboard differently for an executive versus an operations manager?

An executive dashboard would emphasize a small number of high-level KPIs, trends, exceptions, and strategic comparisons. An operations dashboard would generally require more granular metrics, detailed breakdowns, filters, and operational indicators.

12. Deliverables Completed

- Interactive Power BI dashboard file (.pbix).
- Dashboard containing multiple sales-performance visuals.
- Product, regional, and time-based sales analysis.
- Interactive slicer/filter functionality.
- Dashboard reading guidance and analytical documentation.

13. Conclusion

The Sales Dashboard task demonstrates the practical use of Power BI to transform prepared sales data into an interactive reporting solution. The completed dashboard focuses on product, region, and time-based sales analysis while incorporating KPIs and filtering capabilities. The work demonstrates an important part of the data analyst workflow: converting data into a clear visual interface that allows stakeholders to explore performance and obtain information without directly working with the underlying dataset.

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Primary Tool: Microsoft Power BI